

Vishal Kumar Allam

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PROFESSIONAL SUMMARY

- Results-driven **Quality Engineering / Test Automation professional** with **6+ years of experience** leading software testing, automation, test strategy, and quality assurance for enterprise applications in Agile delivery environments.
- Strong hands-on experience designing and implementing scalable automation frameworks using **Selenium, Playwright, Java, C#, .NET, Python, JavaScript, SQL, and Azure SQL** across UI, API, component, integration, and end-to-end testing.
- Extensive expertise in **database and ETL testing**, including SQL-based data validation, Azure SQL, SSIS, source-to-target reconciliation, transformation validation, data integrity checks, and data-driven testing across enterprise data pipelines.
- Experienced implementing **AI-driven and agentic automation solutions** using GitHub Copilot, Codex, Databricks Genie, Python, machine learning-assisted testing, and agentic workflows to improve automation development, analysis, optimization, and test coverage.
- Strong knowledge of QA methodologies including **black-box testing, white-box testing, risk-based testing, functional testing, integration testing, system testing, regression testing, exploratory testing, performance testing, and security testing**.
- Experienced developing comprehensive test plans, test cases, scripts, release estimates, automation strategies, quality metrics, and coverage analysis while performing root-cause analysis and recommending improvements to software design, reliability, and scalability.
- Proven technical leadership collaborating with architects, developers, Product Owners, business stakeholders, and QA teams while mentoring engineers, reviewing test artifacts, driving continuous improvement, and promoting automation and quality-engineering best practices.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming & Automation:** Java, C#, .NET, Python, JavaScript, Selenium WebDriver, Playwright, TestNG, JUnit, Cucumber BDD, Page Object Model
- **AI & Agentic Testing:** AI-Driven Automation, Agentic Workflows, Databricks Genie, GitHub Copilot, Codex, Machine Learning-Assisted Testing, AI Test Optimization, Code Refactoring
- **Database & ETL Testing:** SQL, Azure SQL, PostgreSQL, MySQL, Oracle, ETL Testing, SSIS, Data Validation, Data Transformation Testing, Data Integrity, Source-to-Target Reconciliation
- **Testing Methodologies:** Manual Testing, Functional Testing, Integration Testing, System Testing, Regression Testing, Exploratory Testing, Risk-Based Testing, Black-Box Testing, White-Box Testing
- **Non-Functional Testing:** Performance Testing, Security Testing, Compatibility Testing, API Testing, Database Testing, End-to-End Testing, Data-Driven Testing
- **DevOps & Configuration Management:** Git, GitHub, Jenkins, TeamCity, GitLab CI/CD, Maven, Docker, CI/CD Pipelines, Source Control, Configuration Management
- **Quality Leadership:** Test Strategy, Test Planning, Acceptance Criteria, Release Planning, Estimation, RCA, Test Coverage, Quality Metrics, Defect Management, Mentoring, Agile/Scrum

Test Automation Engineer
PCMI LLC (Full Time)Oct 2024 – Present
Dallas, TX, USA

- Led creation of comprehensive **test strategies, test plans, test cases, and automated test scripts** based on business requirements and acceptance criteria for enterprise SaaS applications across functional, integration, system, regression, and end-to-end testing.
- Designed and maintained scalable automation frameworks using **Selenium WebDriver, Playwright, Java, C#, .NET, Python, JavaScript, TestNG, and reusable Page Object Model components** for UI, API, component, and integration-level validation.
- Implemented **AI-driven and agentic automation solutions** using Python, GitHub Copilot, Codex, and Databricks Genie to assist test generation, script optimization, code refactoring, test-data analysis, defect investigation, and automation coverage improvement.
- Performed extensive database and ETL testing using **SQL, Azure SQL, SSIS, and data-validation utilities**, verifying source-to-target mappings, transformation rules, data integrity, reconciliation, stored data, and downstream application behavior.
- Executed manual and automated **functional, integration, performance, security, black-box, white-box, risk-based, system, regression, and exploratory testing** to identify quality risks throughout the software-development lifecycle.
- Integrated automated tests with Git-based source control and CI/CD pipelines using GitHub, Jenkins, TeamCity, and Maven, supporting repeatable test execution, deployment validation, configuration management, and rapid feedback.
- Analyzed defects and complex production issues through **root-cause analysis**, SQL validation, logs, application traces, and test evidence; recommended design and automation improvements that enhanced system reliability, scalability, and testability.
- Provided technical leadership by reviewing test artifacts, mentoring team members, evaluating test coverage and quality metrics, supporting release planning and estimation, and collaborating with architects, developers, Product Owners, and business stakeholders.

Environment: Selenium WebDriver, Playwright, Java, C#, .NET, Python, JavaScript, SQL, Azure SQL, ETL, SSIS, Databricks Genie, GitHub Copilot, Codex, Agentic Automation, TestNG, Git, Jenkins, TeamCity, Maven, CI/CD, Functional Testing, Integration Testing, Performance Testing, Security Testing, Risk-Based Testing, RCA, Agile Scrum

SDET

Dec 2023 – Sept 2024

Client: CVS

Remote, USA

- Developed enterprise healthcare test plans, scenarios, scripts, and automation strategies from functional and technical requirements, covering application, API, database, data-pipeline, and integration validation across Agile releases.
- Built and enhanced reusable automation frameworks using **Selenium, Playwright, Java, C#, .NET, Python, JavaScript, Cucumber, and TestNG**, enabling component, API, integration, regression, and end-to-end automation.
- Applied **AI-assisted and agentic testing techniques** using GitHub Copilot, Codex, Databricks Genie, and Python-based utilities to accelerate automation creation, optimize existing scripts, analyze failures, generate test scenarios, and improve test coverage.
- Performed comprehensive **SQL, Azure SQL, ETL, SSIS, and database testing**, validating healthcare data transformations, mappings, source-to-target reconciliation, transaction integrity, and data movement across enterprise pipelines.
- Executed functional, integration, regression, system, performance, security, exploratory, black-box, white-box, and risk-based testing while identifying high-impact defects and validating critical healthcare workflows.
- Integrated automation suites with Git, GitHub/GitLab, Jenkins, Maven, and CI/CD pipelines to enable continuous validation, release-quality gates, source-control management, automated regression testing, and deployment verification.
- Performed root-cause analysis across application, API, database, and data layers using logs, SQL queries, test evidence, and system traces, partnering with developers to resolve complex quality and integration issues.
- Collaborated with architects, developers, Product Owners, data engineers, and business partners during sprint planning and release activities while reporting test coverage, defect trends, quality metrics, risks, and release-readiness recommendations.

Environment: Selenium WebDriver, Playwright, Java, C#, .NET, Python, JavaScript, SQL, Azure SQL, ETL, SSIS, Databricks Genie, GitHub Copilot, Codex, AI-Driven Testing, REST APIs, TestNG, Cucumber BDD, Git, GitHub, Jenkins, Maven, CI/CD, Healthcare Data, RCA, Agile Scrum

Software Engineer

Sept 2021 – Dec 2022

Infor

Hyderabad, India

- Created and maintained test plans, test cases, test scripts, and automation solutions for enterprise applications based on requirements, technical specifications, user stories, and acceptance criteria.
- Developed reusable test automation using **Selenium WebDriver, Playwright, Java, C#, JavaScript, Cucumber BDD, and TestNG**, supporting functional, component, integration, system, and regression testing.
- Performed database and ETL validation using **SQL, Azure SQL, relational databases, and ETL workflows**, validating transformation logic, source-to-target mappings, business rules, and data integrity across application layers.
- Executed manual and automated black-box, white-box, exploratory, risk-based, functional, integration, regression, system, performance, and security testing to ensure reliable enterprise software releases.
- Developed SQL queries and backend-validation utilities to verify database transactions, data-processing logic, ETL outputs, API persistence, and application-generated records.
- Integrated automated test suites with Git, Jenkins, Maven, and CI/CD pipelines, supporting source control, continuous integration, automated regression execution, configuration management, and release validation.
- Performed root-cause analysis of complex application and data defects by evaluating logs, SQL records, API responses, system behavior, and test evidence, working with development teams through resolution.
- Collaborated with Software Engineers, Product Managers, QA leads, and business stakeholders while contributing to test estimation, release planning, test coverage analysis, quality reporting, artifact reviews, and continuous process improvement.

Environment: Selenium WebDriver, Playwright, Java, C#, JavaScript, SQL, Azure SQL, ETL, Data Validation, Cucumber BDD, TestNG, Git, Jenkins, Maven, CI/CD, Functional Testing, Integration Testing, Regression Testing, System Testing, Exploratory Testing, Risk-Based Testing, RCA, Agile Scrum

Associate Engineer QA

June 2019 – August 2021

Virtusa

Hyderabad, India

- Developed and executed detailed test cases, test plans, test scripts, and validation scenarios based on business requirements and acceptance criteria for enterprise web, mobile, API, and backend systems.
- Built and maintained automated regression frameworks using **Selenium WebDriver, Java, C#, Cucumber BDD, TestNG, Appium, and reusable automation patterns** for functional and system-level validation.
- Performed SQL, database, and ETL testing to validate application transactions, source-to-target mappings, transformation rules, business data, and data integrity across integrated enterprise systems.
- Executed comprehensive **manual, functional, integration, performance, security, system, regression, exploratory, black-box, white-box, and risk-based testing** throughout Agile development cycles.
- Developed API and integration-validation scenarios covering REST services, request/response payloads, downstream services, database updates, and cross-system business workflows.
- Integrated automated test suites with Git, Jenkins, Maven, and CI/CD workflows to improve automation efficiency, repeatable regression execution, build validation, source-control practices, and software release quality.
- Investigated and tracked complex defects through root-cause analysis, logs, SQL validation, API responses, test evidence, and collaboration with development teams, ensuring issues were resolved before release.
- Participated in Agile sprint planning, test estimation, defect triage, release planning, test-artifact reviews, quality-metric reporting, and continuous improvement while sharing automation and QA best practices across the team.

Environment: Selenium WebDriver, Java, C#, Cucumber BDD, TestNG, Appium, SQL, ETL, Database Testing, REST APIs, Git, Jenkins, Maven, CI/CD, Manual Testing, Functional Testing, Integration Testing, Performance Testing, Security Testing, Black-Box Testing, White-Box Testing, Risk-Based Testing, RCA, Agile Scrum

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.